

Electricity Fraud Detection System Using Smart Meter Data

Aishat Saleh Muhammad · ALT/SOD/025/3513

AltSchool Africa – Data Science · Karatu 2025 Third Semester
Capstone

Problem Statement

The Challenge

Electricity theft (meter tampering, illegal connections, meter bypassing) is one of the largest sources of Non-Technical Losses (NTL) for Nigerian Distribution Companies (DisCos).

Consequences include:

- Significant revenue losses
- Transformer overload & grid instability
- Higher maintenance costs
- Degraded service for paying customers

Problem with Current Approach

- Complaint-driven investigations
- Random field inspections
- Slow, expensive fraud detection
- Near-zero link to actual fraud cases

Project Goal

Develop a machine learning system to identify fraudulent consumption patterns using smart meter telemetry — enabling DisCos to move from reactive inspections to proactive, data-driven fraud detection.

Dataset & Methodology

Dataset Overview

15,000

smart meter
records

14,849

records after
cleaning

500

customers
monitored

5

Nigerian DisCos
covered

30 days

observation period

DisCos Covered

Abuja AEDC

Kano KEDCO

Eko DisCo

Ikeja Electric

Jos JED

Methodology

1. Business Understanding
2. Data Cleaning
3. EDA
4. Feature Engineering
5. Statistical Analysis
6. Machine Learning
7. Dashboard Development

Data Preparation & Feature Engineering

Data Cleaning

- Removed 151 records with missing DisCo (non-imputable)
- Replaced sensor error codes (-99) with NaN
- Imputed missing values using tariff-band median
- Retained 672 voltage outliers as fraud signals
- Recalculated revenue after imputation
- Final dataset: 0 missing values

Feature Engineering

All 5 features computed after chronological split, independently on each partition — no future data in training window.

Consumption_7d_mean

7-day rolling average per meter

Consumption_Drop_From_Mean

Deviation from baseline (bypass signal)

Voltage_7d_std

Rolling voltage instability

Leakage_7d_max

Worst leakage spike in 7 days

Consumption_Lag_1

Yesterday's consumption

Exploratory Data Analysis

Avg daily consumption

19.20

kWh

Avg voltage

220.84

Volts

Max current leakage

0.90

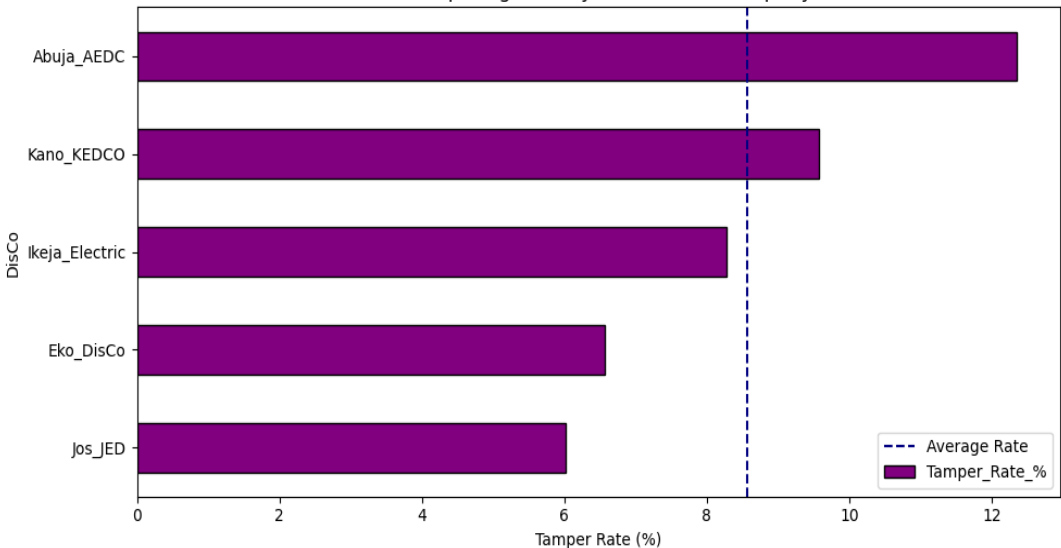
Amps

Overall fraud rate

8.59

%

Meter Tampering Rate by Distribution Company (%)



Fraud Distribution

Normal meters

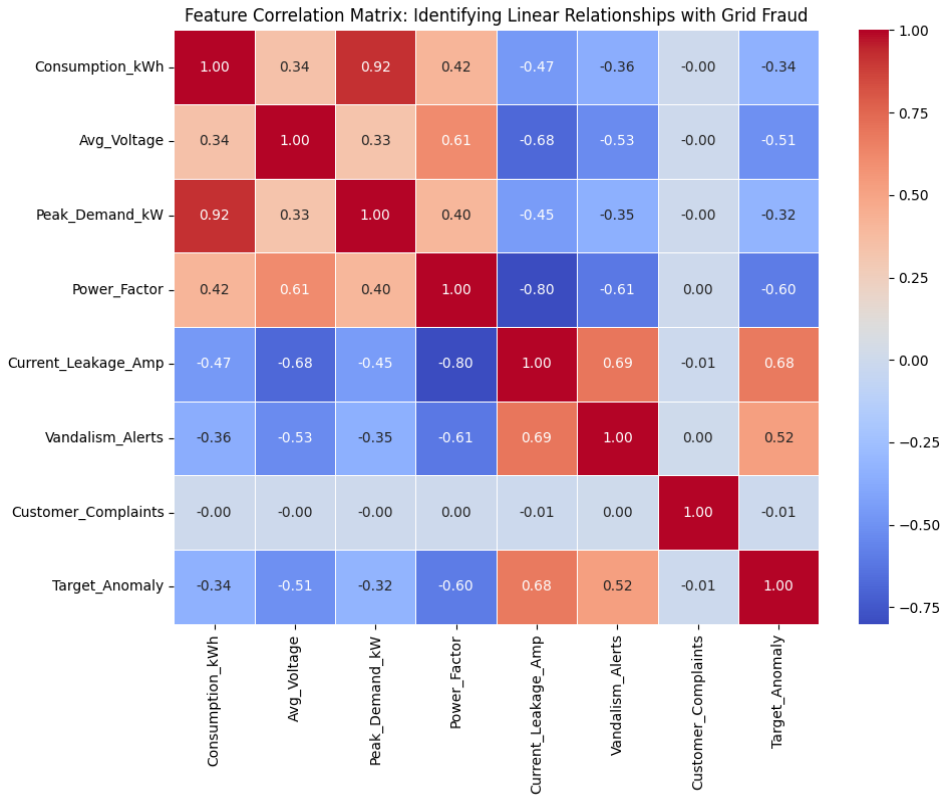
13,574

Fraud meters

1,275

Key insight: Fraud is not evenly distributed. Abuja AEDC has 2× the fraud rate of Jos JED — targeting high-risk DisCos improves inspection efficiency significantly.

Key Fraud Indicators



What the signals mean

Current leakage +0.68

Physical wires bypassing the meter cause electricity to leak into grounding systems — the strongest fraud fingerprint.

Power factor -0.60

Circuit efficiency drops when internal meter connections are bridged.

Avg voltage -0.51

Bypassing creates localised voltage depression detectable by telemetry.

Customer complaints ≈ 0.00

Thieves do not self-report. Complaint-driven inspections are misdirected.

Machine Learning Model

Algorithm: Random Forest Classifier

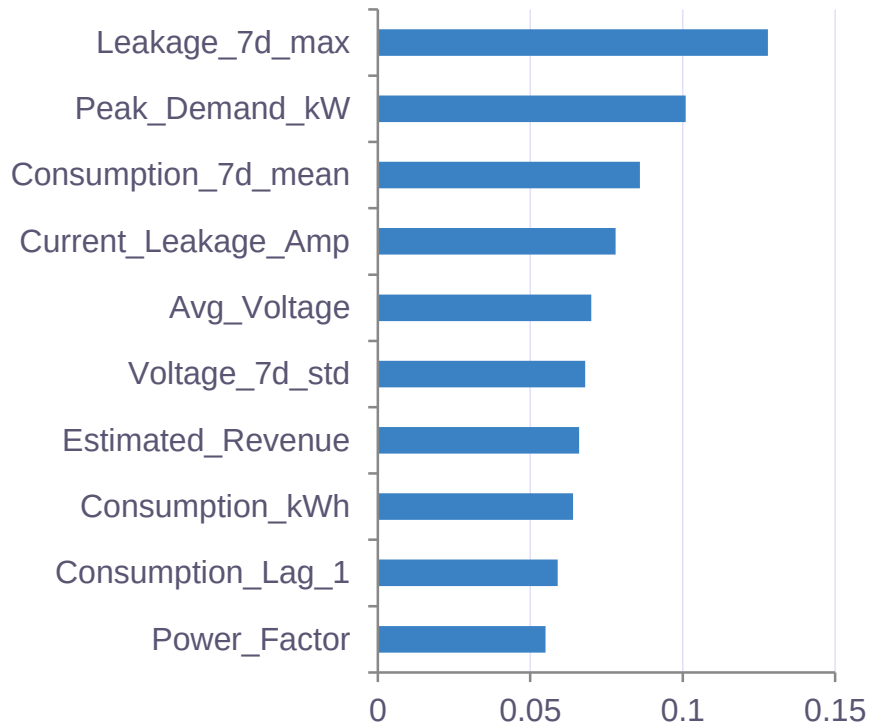
Why Random Forest?

- Handles non-linear relationships effectively
- Robust to noisy sensor data
- Performs well with imbalanced datasets (class_weight='balanced')
- Provides interpretable feature importance rankings

Top predictors (feature importance)

1. Leakage_7d_max
2. Peak_Demand_kW
3. Consumption_7d_mean

Top 10 Feature Importances



Model Performance

Accuracy

99.97

%

Precision (fraud)

~100

%

Recall (fraud)

100

%

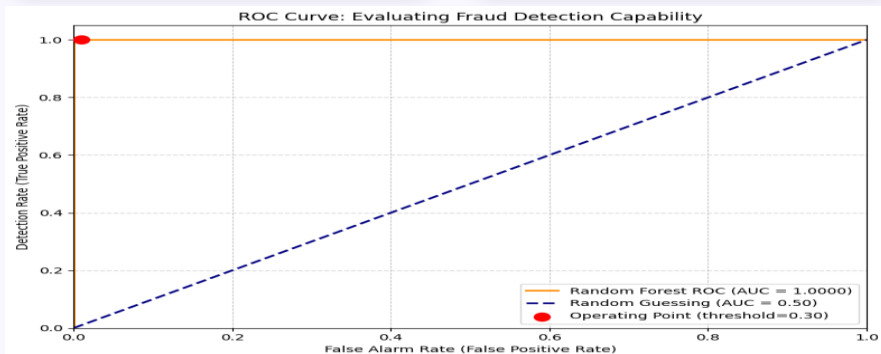
ROC-AUC

1.00

score

Confusion matrix (test set: May 25–30)

	Predicted Normal	Predicted Fraud
Actual Normal	2,716 ✓	0
Actual Fraud	1	254 ✓



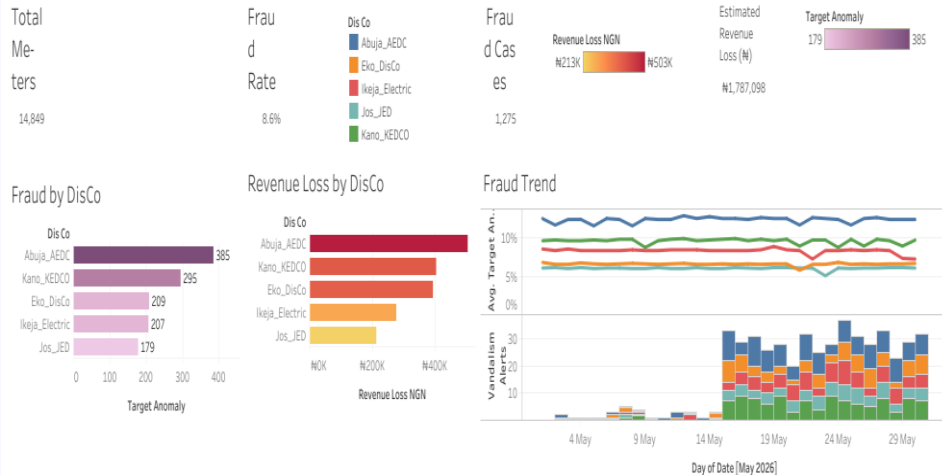
Note on AUC = 1.00

These scores reflect strong physical fraud signals — current leakage spikes and voltage drops are direct engineering consequences of meter tampering, not statistical coincidences. Data leakage was prevented: rolling features were computed after a chronological split, independently on each partition. Real-world performance will be lower due to sensor noise, missing telemetry days, and concept drift; periodic retraining is recommended.

Tableau Dashboard Insights

Nigerian Grid Fraud Detection Dashboard

Summary: During May 2026, data tracking across the Distribution Companies (DisCos) identified a **8.6% Fraud Rate** across 14,849 meters, resulting in an estimated revenue loss of **₦1.78M**. Vandalism alerts and target anomalies peaked heavily in the latter half of the month.



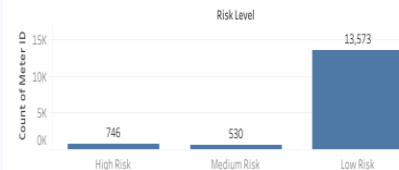
Key Insights

- Fraud concentration by DisCo:** Abuja_AEDC recorded the highest number of fraud cases (385) and the highest revenue loss. Kano_KEDCO ranked second in both fraud cases and revenue loss. These findings suggest that fraud prevention efforts should be prioritized in these high-risk distribution companies.
- Financial impact:** The total estimated revenue loss exceeds ₦1.7 million. A significant portion of this loss is concentrated within a few DisCos, indicating opportunities for targeted interventions.
- Fraud trend over time:** Fraud incidents and vandalism alerts were observed throughout May 2026. This suggests that meter tampering is an ongoing operational challenge that requires continuous monitoring and preventive measures.

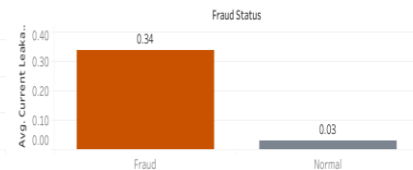
Fraud Risk Analysis

This dashboard highlights fraud indicators and identifies high-risk meters requiring further investigation.

Risk Distribution



Leakage Current by Fraud Status



Vandalism Alerts by Fraud Status



Revenue Loss by Risk Level



Key Insights

- Risk distribution:** Most meters were classified as Low Risk. However, because they represent a large proportion of the meter population, they also contribute substantially to the overall revenue loss. This highlights the need to monitor both high-risk and low-risk groups.
- Leakage current as a fraud indicator:** Fraudulent meters showed a noticeably higher average leakage current (0.34 A) compared to normal meters (0.03 A). This suggests that leakage current may be a useful indicator for identifying potential meter tampering.
- Vandalism alerts and fraud:** Fraudulent meters recorded higher average vandalism alerts than normal meters. This relationship suggests that physical tampering events may be associated with increased fraud risk.

Recommendations

- Prioritize inspections in Abuja_AEDC and Kano_KEDCO. Use leakage current and vandalism alerts as early warning indicators...

Key insight

Medium-risk meters account for **₦1.46M (82%)** of total at-risk revenue. Monitoring strategy cannot focus only on the 746 high-risk flagged meters — the medium tier requires parallel attention to recover the majority of projected losses.

Recommendations & Conclusion

1. Prioritise Abuja AEDC & Kano KEDCO

These DisCos account for the majority of fraud cases and revenue loss — allocate field resources here first.

2. Monitor Tariff Bands C & D

12.01% and 11.04% fraud rates respectively — schedule more frequent meter audits for these segments.

3. Deploy tamper-proof meters with vandalism sensors

One alert = 87.45% fraud probability — sensors are a high-ROI investment.

4. Replace complaints with ML risk scoring

Complaints correlate ≈ 0.00 with fraud — redirect those inspection hours to model-flagged meters.

5. Retrain the model quarterly

Real-world performance will drift; fresh labelled audit data is required to maintain accuracy.

Conclusion

Machine learning enables Nigerian DisCos to shift from reactive inspections to proactive, data-driven fraud detection — identifying 1,275 fraud cases and ₦1.78M at-risk revenue across 14,849 meters in a 30-day period.

Thank You

Questions & Discussion

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